INCH-POUND

MIL-S-46897B(MI)
5 October 1990
SUPERSEDING
MIL-S-46897A(MI)
29 April 1983

MILITARY SPECIFICATION

SEALING COMPOUND, POLYURETHANE FOAM

This specification is approved for use by the U.S. Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of polyurethane foam sealing compound used as a nozzle seal in rocket motors.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Box, Wood, Nailed and Lock Corner
PPP-C-96	Can, Metal, 28 Gage and Lighter

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8030

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STANDARDS

MILITARY

MIL-STD-129 Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

American Society for Testing and Materials

ASTM D 1622	Test for Apparent Density of Rigid Cellular Plastics
ASTM D 1638	Testing Urethane Foam Isocyanate Raw Materials
ASTM D 2849	Urethane Foam Polyol Raw Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption is obtained.

3. REQUIREMENTS

3.1 Materials. The material shall be a rigid, closed-cell polyurethane foam, and shall be a self-curing, two-component system, consisting of a polyol component and an isocyanate component.

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3.2 Chemical and physical properties. The chemical and physical properties shall be as specified in Table I.

Table I. Chemical and physical properties.

Property	Minimum	Maximum	Test Method
Polyol component			
Specified gravity	1.22	1.30	4.4.2
Hydroxyl number	455	515	4.4.3
Isocyanate component			
Specific gravity	1.12	1.20	4.4.2
Isocyanate content (percent)	32	38	4.4.4
Foam			
Bulk density, lb/ft^3	4.5	7.5	4.4.5

The use of freon, chlorofluorocarbons, or other compounds, which are restricted because of their effects on the ozone layer, are forbidden.

3.3 Workmanship. The sealing compound components shall be uniform in quality and free from impurities and other defects that could adversely affect their use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the

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specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as quality conformance inspection (see 4.3).

4.3 Quality conformance inspection.

4.3.1 Inspection lot. An inspection lot shall consist of material from one batch or run subjected to the same processing operations and conditions by the same manufacturer, and presented for inspection at one time.

4.3.2 Sampling. Sampling for visual examination shall be 100 percent of the lot. Sampling for all other examinations and tests shall be a representative sample selected at random. Failure of the sample to meet any requirement of this specification shall cause rejection of the lot.

4.3.3 Procedure. Samples taken in accordance with 4.3.2 shall be subjected to the examinations and tests as listed in Table II.

Table II. Examinations and tests.

Examination	Test Paragraph	Requirement Paragraph
Visual	4.4.1	3.1, 3.3, 5
Specific gravity	4.4.2	3.2
Hydroxyl number	4.4.3	3.2
Isocyanate content	4.4.4	3.2
Bulk density	4.4.5	3.2

4.4 Inspection methods. The inspection methods shall be as follows:

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4.4.1 Visual examination. Visual examination of the samples specified in 4.3.2 shall be conducted to determine compliance with 3.1, 3.3, and 5.

4.4.2 Specific gravity. The specific gravity of the polyol component materials shall be determined in accordance with ASTM D 2849 and the specific gravity of the isocyanate component materials shall be determined in accordance with ASTM D 1683.

4.4.3 Hydroxyl number. The hydroxyl number of the polyol component shall be determined in accordance with ASTM D 2849.

4.4.4 Isocyanate content. The isocyanate content of the curing agent shall be determined in accordance with ASTM D 1638.

4.4.5 Bulk density. The bulk density of the cured foam shall be determined in accordance with ASTM D 1622.

5. PACKAGING

5.1 Preservation. Preservation, packaging, packing, unitization, and marking shall be Level A or B (see 6.2).

5.1.1 Level A. Polyurethane shall be packaged in containers conforming to PPP-C-96 or a commercial type container suitable for polyurethane foam.

5.1.2 Level B. Same as Level A.

5.2 Packing.

5.2.1 Level A. Packaged polyurethane shall be packed in snug-fitting containers conforming to PPP-B-601, overseas type or PPP-B-621 Class 2 as applicable.

5.2.2 Level B. Same as Level A.

5.3 Marking.

5.3.1 Level A and B. Marking shall be in accordance with MIL-STD-129. Additional marking shall include, but not be limited to the following:

a. Manufacturer's name and location.

b. Material trade name.

c. Net weight.

d. Lot number, batch number, and date of manufacture.

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e. Number and revision letter of this specification.

f. Shelf life or storage limitations.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.

6.1 Intended use. This type of sealing compound is intended for use as a nozzle seal in rocket motors.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1 and 2.2).

c. Minimum lot size, if applicable.

d. Place of delivery.

e. Size of container, or foam sheet.

f. Place of inspection.

g. Disposition of test data.

h. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Subject term (key word) listing.

Gravity
Hydroxyl
Isocyanate
Nozzle
Resin

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6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - MI

Preparing activity:
Army - MI

Review activities:
Army - MR

Project No. 8030-A125

Civilian Coordinating Activity:
GSA - FSS

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